Operational Diagnostics Report

Nairobi Depot Throughput Shortfall — Week 5 Investigation

Prepared by: Analyst **Date:** 05 September 2026 **Scope:** Full fleet, calendar year 2026

Context

Fleet-wide daily production output ("throughput") has been running below target for most of 2026. This report investigates 2,920 daily production records across our four depots — Nairobi, Mombasa, Kisumu, and Eldoret — and 13 production machines, covering every shift for the full calendar year, to identify exactly where the shortfall is coming from and what is driving it.

Insight: What We Found

The problem is not fleet-wide — it is one machine. Machine **NBI-P03** at the Nairobi depot ran normally through January, averaging about 1,028 barrels/day, in line with every other machine. Starting **February 1, 2026**, its output dropped to roughly **618 barrels/day — a 40% loss of capacity — and it never recovered** for the remaining eleven months of the year. The chart below shows this clearly: the red points (NBI-P03) split away from the rest of the fleet (gray) at a single, identifiable point in time.

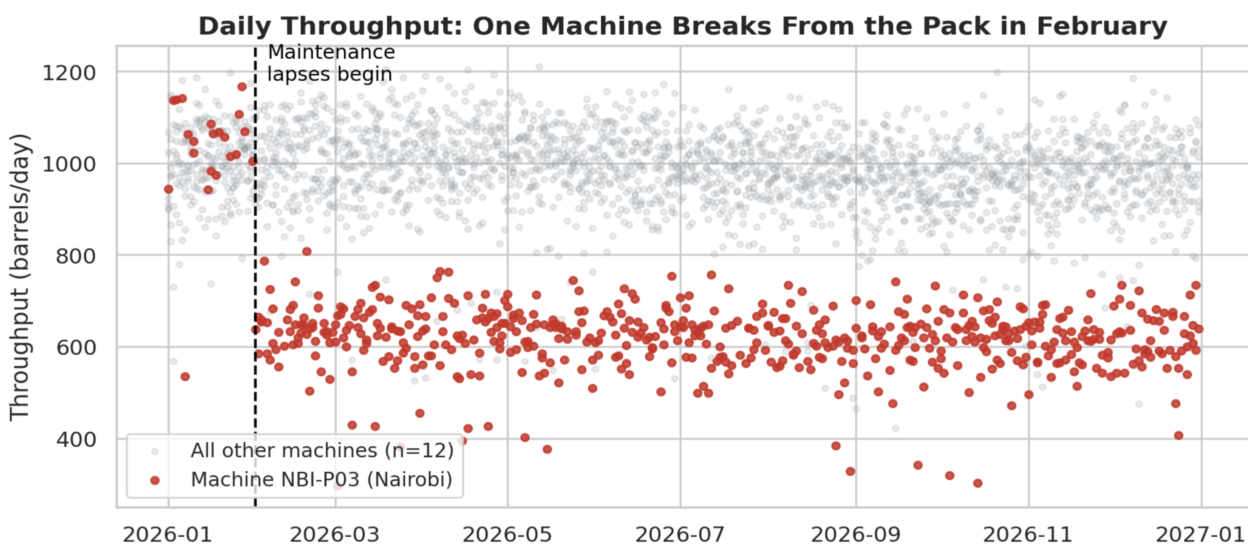

Figure 1. Daily throughput for every machine in the fleet. NBI-P03 (red) breaks away from the rest of the fleet (gray) on February 1 and stays low for the rest of the year.

We traced this back to the logged reason for each low-output day, and the pattern is stark: NBI-P03's low-output days are almost all tagged as **"missed maintenance"** — meaning a scheduled service was skipped or delayed, allowing the equipment to keep running in a degraded state instead of being restored to full capacity. Across the *entire fleet*, missed-maintenance incidents account for **88% of all throughput lost** this year — far more than equipment faults or operator error combined (Figure 2) — and NBI-P03 alone is responsible for **about 86% of that total loss**, even though it is only 1 of 13 machines.

We also checked two other possible explanations before settling on this conclusion: weather (temperature) and electrical supply variance (voltage). Neither showed any meaningful relationship to the drop in output, and the three other machines at the same Nairobi depot performed normally all year — which rules out a site-wide or environmental cause and confirms this is isolated to the one machine.

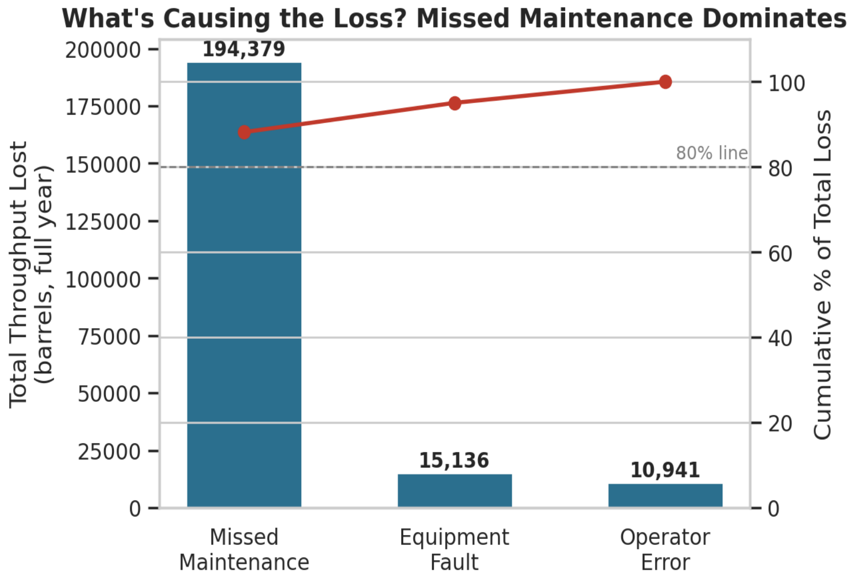

Figure 2. Total throughput lost in 2026, by cause. Missed maintenance alone accounts for 88% of all lost output fleet-wide — well past the 80% line that marks a dominant cause.

Action: What We Recommend

1

Fix NBI-P03 immediately

Send a maintenance crew to inspect and service NBI-P03 as a stand-alone priority repair — not part of the next routine fleet-wide maintenance cycle. Every week this waits costs roughly 2,900 barrels of lost output.

2

Find out why the service was missed

This machine's scheduled maintenance was skipped or delayed starting in late January and was never caught for eleven months. Before closing this out, confirm whether the cause was a scheduling gap, a parts shortage, or a staffing issue at the Nairobi depot — and fix that process, not just this one machine.

3

Add a monthly throughput check per machine

A simple monthly comparison of each machine's output against its own historical baseline would have flagged this drop within the first month, instead of a year later. This is a low-cost check that prevents the next NBI-P03 from going unnoticed for so long.

Bottom line: this is a single-asset maintenance failure, not a fleet-wide problem — a targeted repair on NBI-P03, paired with a look at why its maintenance schedule slipped through the cracks, should recover the bulk of this year's lost output.